25 Structured Exam-Style Questions **Name:** _____ **Date:** _____ **Score:** _____ / 25

Vector Spaces & The Four Fundamental Subspaces Practice Exercises

1 MCQ

Find $\|\mathbf{u}\|$ for $\mathbf{u} = (-9, -12)$.

A 21
B 14
C 15
D 225

2 MCQ

A 3×3 symmetric matrix Q_k has eigenvalues $\lambda_1 = 12$, $\lambda_2 = 3$, and $\lambda_3 = 14$. Determine the trace $\text{tr}(Q_k)$.

A 26
B 32
C 29
D 41

3 MCQ

A linear transformation $T : \mathrm{R}^2 \to \mathrm{R}^2$ is defined by $T(\mathbf{x}) = A\mathbf{x}$, where $A = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$. If $\mathbf{v} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$, find $T(\mathbf{v})$.

A $\begin{pmatrix} 4 \\ 7 \end{pmatrix}$
B $\begin{pmatrix} 3 \\ 7 \end{pmatrix}$
C $\begin{pmatrix} 4 \\ 5 \end{pmatrix}$
D $\begin{pmatrix} 5 \\ 7 \end{pmatrix}$

4 MCQ

Consider the matrix $A = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$. If a point $P = (x, y)$ is transformed by A to $P' = (x', y')$, such that $\begin{pmatrix} x' \\ y' \end{pmatrix} = A \begin{pmatrix} x \\ y \end{pmatrix}$, what is the image of the point $(3, 1)$ under this transformation?

A (7, 6)
B (6, 7)
C (8, 5)
D (5, 8)

5 MCQ

Consider two vectors $\mathbf{v}_1 = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$ and $\mathbf{v}_2 = \begin{pmatrix} -4 \\ 3 \end{pmatrix}$. What is the dot product $\mathbf{v}_1 \cdot \mathbf{v}_2$?

A 1
B 0
C -7
D 25

6 MCQ

Consider the following system of linear equations represented by the augmented matrix:

$$\left(\begin{array}{cc|c} 1 & 2 & 5 \\ 2 & -1 & 0 \end{array}\right)$$

. Use row reduction to transform the matrix into row-echelon form and determine the unique solution (x, y) for the system. What is the value of x?

A 1

B 2

C 3

D 4

7 MCQ

Consider the vector $\mathbf{v} = \begin{pmatrix} 5 \\ -12 \end{pmatrix}$. Calculate the magnitude of $\mathbf{v}$, denoted as $\|\mathbf{v}\|$.

A 13

B 17

C 7

D 169

8 MCQ

Consider the vector $\mathbf{v} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$. If this vector is plotted on a standard xy-plane, it represents a directed line segment from the origin to the point $(3, 4)$. What is the geometric interpretation of the length of this vector?

A The vector's direction in the xy-plane.

B The angle the vector makes with the positive x-axis.

C The distance of the point (3,4) from the origin.

D The sum of the components of the vector.

9 MCQ

Consider the linear transformation $T : \mathrm{R}^3 \to \mathrm{R}^3$ defined by $T(\mathbf{x}) = A\mathbf{x}$, where $A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$. Let V be the image of T, i.e., $V = \mathrm{Im}(T)$. V is a subspace of R^3. Find the dimension of the orthogonal complement of V, denoted by $V^{\perp}$.

A 2

B 0

C 3

D 1

10 MCQ

Consider the vector $\mathbf{v} = \begin{pmatrix} -3 \\ 4 \end{pmatrix}$ in R^2. What is the magnitude (or norm) of this vector?

A 5

B 7

C 1

D 25

11 MCQ

Consider the linear transformation $T : \mathrm{R}^2 \to \mathrm{R}^2$ represented by the matrix $A = \begin{pmatrix} 1 & 2 \\ k & 4 \end{pmatrix}$. The image of the vector $\mathbf{v} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ under this transformation is $T(\mathbf{v}) = \begin{pmatrix} 3 \\ 5 \end{pmatrix}$. Determine the value of the parameter k.

A 2

B 1

C 3

D 4

12 MCQ

Consider a linear transformation $T : \mathrm{R}^3 \to \mathrm{R}^2$ defined by $T(x, y, z) = (2x - y + z, x + 3y - 2z)$. Let V be the image of T, i.e., $V = \mathrm{Im}(T)$. Furthermore, let W be the subspace of R^2 defined by $W = \{(a, b) \in \mathrm{R}^2 \mid a + b = 0\}$. Find the dimension of the intersection of V and W, i.e., $\dim(V \cap W)$.

A 1

B 0

C 2

D Cannot be determined

13 MCQ

Consider the vector space R^2 with the standard dot product. Let $\mathbf{u} = (3, 4)$ and $\mathbf{v} = (2, -1)$. Calculate the dot product $\mathbf{u} \cdot \mathbf{v}$.

A 10

B -2

C 6

D 2

14 MCQ

Consider the vector $\mathbf{v} = (5 \ \ -12)$. Calculate the magnitude of $\mathbf{v}$, denoted as $\|\mathbf{v}\|$.

A 7

B 169

C 13

D 17

15 MCQ

A linear map $S : \mathrm{R}^{20} \to \mathrm{R}^{28}$ has rank $\mathrm{rank}(S) = 15$. Determine the nullity $\dim(\ker S)$.

A 0

B 20

C 8

D 5

16 MCQ

A planar linear transformation S scales b by 18 and y by 15 with matrix $W = \begin{pmatrix} 18 & 0 \\ 0 & 15 \end{pmatrix}$. Determine the matrix trace $\operatorname{tr}(W)$.

A 33

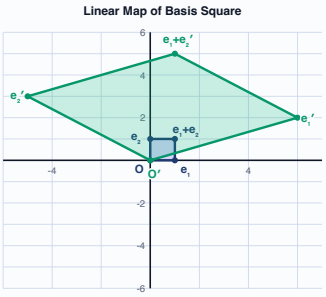
B 51

C 36

D 18

17 MCQ

For $A = \begin{bmatrix} 6 & -5 \\ 2 & 3 \end{bmatrix}$, find entry $(-3A)_{22}$.



A 3

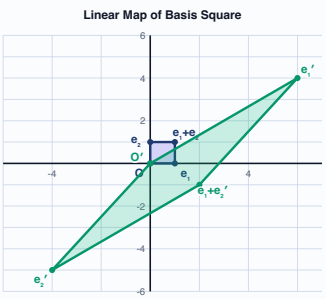
B 9

C -9

D -8

18 MCQ

For $A = \begin{bmatrix} 6 & -4 \\ 4 & -5 \end{bmatrix}$, find entry $(-2A)_{22}$.



A -5

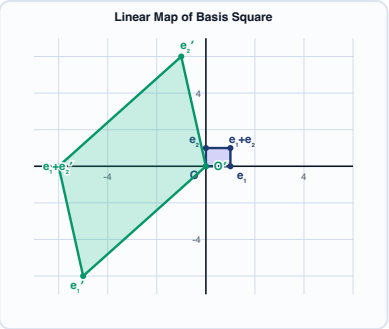
B -10

C 10

D 11

19 MCQ

For $A = \begin{bmatrix} -5 & -1 \\ -6 & 6 \end{bmatrix}$, $B = \begin{bmatrix} -5 & -1 \\ -1 & -4 \end{bmatrix}$, find entry $(A+B)_{12}$.



A -2

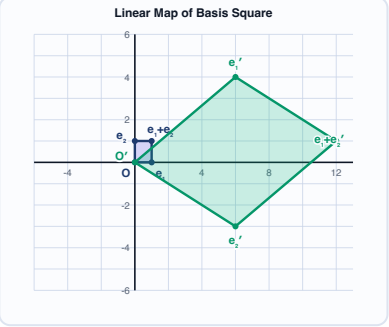
B -5

C -1

D -4

20 MCQ

For $A = \begin{bmatrix} 6 & 6 \\ 4 & -3 \end{bmatrix}$, find entry $(3A)_{21}$.



A 4

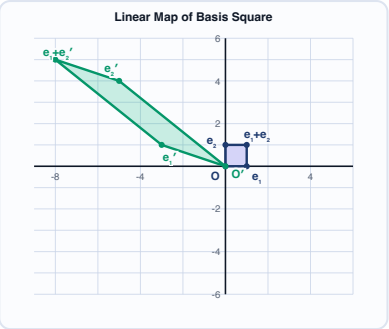
B -12

C 12

D 13

21 MCQ

For $A = \begin{bmatrix} -3 & -5 \\ 1 & 4 \end{bmatrix}$, $B = \begin{bmatrix} -1 & -2 \\ 4 & 6 \end{bmatrix}$, find entry $(A+B)_{12}$.



A -5

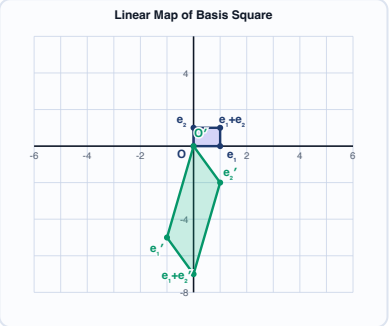
B -7

C -2

D -6

22 MCQ

For $A = \begin{bmatrix} -1 & 1 \\ -5 & -2 \end{bmatrix}$, $B = \begin{bmatrix} -4 & -5 \\ -2 & -1 \end{bmatrix}$, find entry $(A+B)_{22}$.



A -2

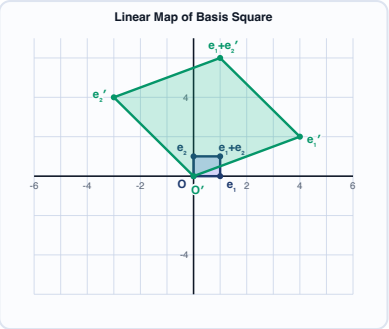
B -1

C -5

D -3

23 MCQ

For $A = \begin{bmatrix} 4 & -3 \\ 2 & 4 \end{bmatrix}$, $B = \begin{bmatrix} -3 & -3 \\ 4 & -1 \end{bmatrix}$, find row 1 of AB.



A [-24, -9]

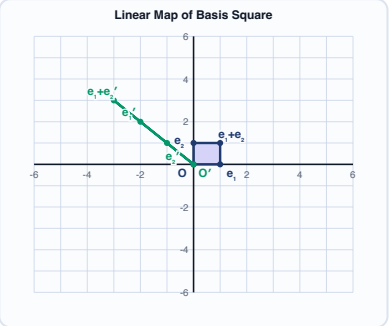
B [-18, -3]

C [-9, -24]

D [-12, 3]

24 MCQ

For $A = \begin{bmatrix} -2 & -1 \\ 2 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 4 & -1 \\ -3 & 4 \end{bmatrix}$, find row 2 of AB.



A [14, 7]

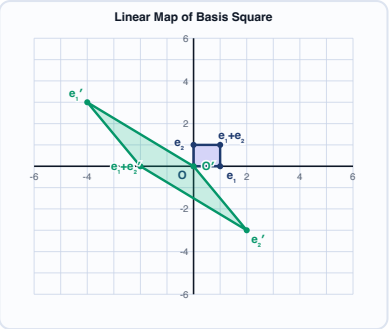
B [5, 2]

C [2, 5]

D [8, 4]

25 MCQ

For $A = \begin{bmatrix} -4 & 2 \\ 3 & -3 \end{bmatrix}$, $B = \begin{bmatrix} -1 & -2 \\ 2 & -2 \end{bmatrix}$, find row 1 of AB.



A [8, 4]

B [4, 8]

C [4, -4]

D [-2, 4]